RACE
Regular Analysis Through Continuous Exercise

Electromagnetic Induction (EMI) : RACE-01

1. The flux linked with a coil at any instant 't' is given by $\phi = 10t^2 - 50t + 250$. The induced emf at $t = 3\ s$ is

(A) 190 V (B) $-190V$ (C) $-10\ V$ (D) $10V$

PEM183

2. A rectangular loop of length 2.5 m and width 2 m is placed at 60° to a magnetic field of $4T$. The loop is removed from the field in 10 sec. The average emf induced in the loop during this time is

(A) $-2V$ (B) $+2V$ (C) $+1V$ (D) $-1V$

PEM184

3. A coil is placed perpendicular to a magnetic field of 5000 T. When the field is changed to 3000 T in 2s, an induced emf of 22 V is produced in the coil. If the diameter of the coil is 0.02 m, then the number of turns in the coil is :

(A) 7 (B) 70 (C) 35 (D) 140

PEM185

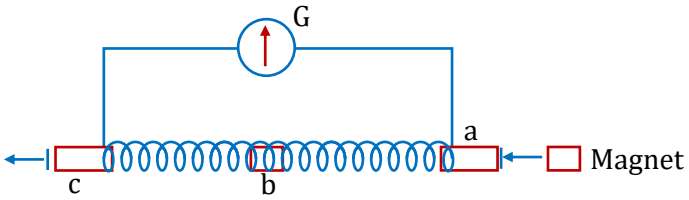
4. A coil of 200 turns and area 0.20 m^2 is rotated at half a revolution per second and is placed in uniform magnetic field of $0.01T$ perpendicular to axis of rotation of the coil. The maximum voltage generated in the coil is $\frac{2\pi}{\beta}$ volt. The value of β is _____.

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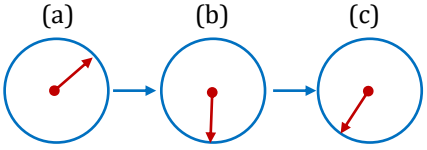
(A) 4 (B) 5 (C) 6 (D) 7

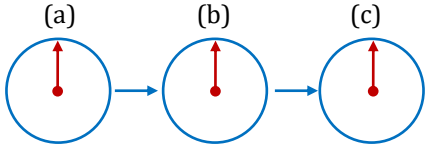
PEM186

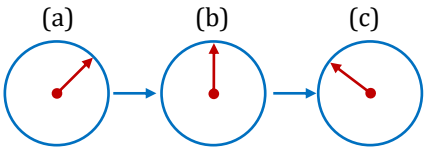
5. A small bar magnet is moved through a coil at constant speed from one end to the other. Which of the following series of observations will be seen on the galvanometer G attached across the coil ?

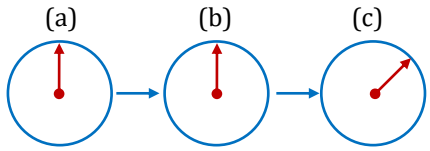


Three positions shown describe : (a) the magnet's entry (b) magnet is completely inside and (c) magnet's exit.

(A) 

(B) 

(C) 

(D) 

PEM187

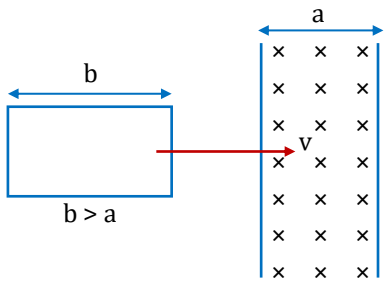
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6. A conducting circular loop is placed in $X-Y$ plane in presence of magnetic field $\vec{B} = \left(3t^3\hat{j} + 3t^2\hat{k}\right)$ in SI unit. If the radius of the loop is $1m$, the induced emf in the loop, at time, $t = 2s$ is $n\pi V$. The value of n is _____.

(A) 10 (B) 11 (C) 12 (D) 13

PEM188

7. In the given arrangement, the loop is moved with constant velocity v in a uniform magnetic field B in a restricted region of width a. The time for which the emf is induced in the circuit is:



(A) $\frac{2b}{v}$ (B) $\frac{2a}{v}$ (C) $\frac{(a+b)}{v}$ (D) $\frac{2(a-b)}{v}$

PEM189

8. A solid metal cube of edge length 2 cm is moving in a positive y direction at a constant speed of 6 m/s. There is a uniform magnetic field of 0.1 T in the positive z-direction. The potential difference between the two faces of the cube perpendicular to the x-axis, is :

(A) 6 mV (B) 1 mV (C) 12 mV (D) 2 mV

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PEM190

9. A horizontal straight wire 5 m long extending from east to west falling freely at right angle to horizontal component of earth's magnetic field 0.60×10^{-4} Wbm^{-2}. The instantaneous value of emf induced in the wire when its velocity is 10 ms^{-1} is _____ $\times 10^{-3}$ V.

(A) 1 (B) 6 (C) 3 (D) 9

PEM191

10. A rectangular loop of sides 12 cm and 5 cm, with its sides parallel to the x-axis and y-axis respectively moves with a velocity of 5 cm/s in the positive x-axis direction, in a space containing a variable magnetic field in the positive z direction. The field has a gradient of 10^{-3} T/cm along the negative x direction and it is decreasing with time at the rate of 10^{-3} T/s. If the resistance of the loop is 6 $m\Omega$, the power dissipated by the loop as heat is _____ $\times 10^{-9}$ W.

(A) 216 (B) 210 (C) 218 (D) 220

PEM192

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	B	B	C	C	B	C	C	A

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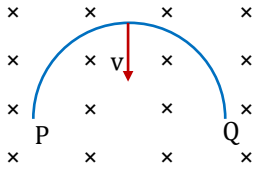
Electromagnetic Induction (EMI) : RACE-02

1. Two concentric circular coils, C_1 and C_2, are placed in the XY plane. C_1 has 500 turns, and a radius of 1 cm. C_2 has 200 turns and radius of 20 cm. C_2 carries a time dependent current $I(t) = (5t^2 - 2t + 3)\ A$ where t is in s. The emf induced in C_1 (in mV), at the instant $t = 1s$ is $\frac{4}{x}$. The value of x is _____.

(A) 2 (B) 3 (C) 4 (D) 5

PEM193

2. A semicircle loop PQ of radius 'R' is moved with velocity 'v' in transverse magnetic field as shown in figure. The value of induced emf. at the end of loop is :–



(A) $Bv\ (\pi r)$, end 'P' at high potential (B) 2 BRv, end P at high potential

(C) 2 BRv, end Q at high potential (D) $B\frac{\pi R^2}{2}v$, end P at high potential

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PEM194

3. A planar loop of wire rotates in a uniform magnetic field. Initially, at $t = 0$, the plane of the loop is perpendicular to the magnetic field. If it rotates with a period of 10s about an axis in its plane then the magnitude of induced emf will be maximum and minimum, respectively at :

(A) 2.5s and 7.5s (B) 5.0s and 7.5s (C) 5.0s and 10.0s (D) 2.5s and 5.0s

PEM195

4. A long solenoid of radius R carries a time (t)-dependent current $I(t) = I_0t(1 - t)$. A ring of radius $2R$ is placed coaxially near its middle. During the time interval $0 \le t \le 1$, the induced current (I_R) and the induced EMF(V_R) in the ring change as:

(A) At $t = 0.5$ direction of I_R reverses and V_R is zero.

(B) Direction of I_R remains unchanged, and V_R is zero at $t = 0.25$.

(C) Direction of I_R remains unchanged, and V_R is maximum at $t = 0.5$.

(D) At $t = 0.25$ direction of I_R reverses and V_R is maximum.

PEM196

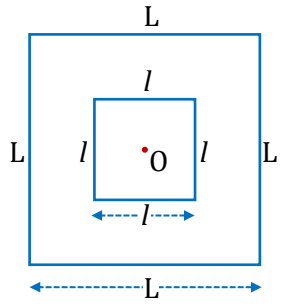
5. The self induced emf of a coil is 25 volts. When the current in it is changed at uniform rate from 10 A to 25 A in 1s, the change in the energy of the inductance is:

(A) 437.5 J (B) 637.5 J (C) 740 J (D) 540 J

PEM197

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6. A small square loop of wire of side l is placed inside a large square loop of wire L ($L >> l$). Both loops are coplanar and their centres coincide at point O as shown in figure. The mutual inductance of the system is :



(A) $\frac{2\sqrt{2}\mu_0 L^2}{\pi\ell}$ (B) $\frac{\mu_0\ell^2}{2\sqrt{2}\pi L}$ (C) $\frac{2\sqrt{2}\mu_0\ell^2}{\pi L}$ (D) $\frac{\mu_0 L^2}{2\sqrt{2}\pi\ell}$

PEM198

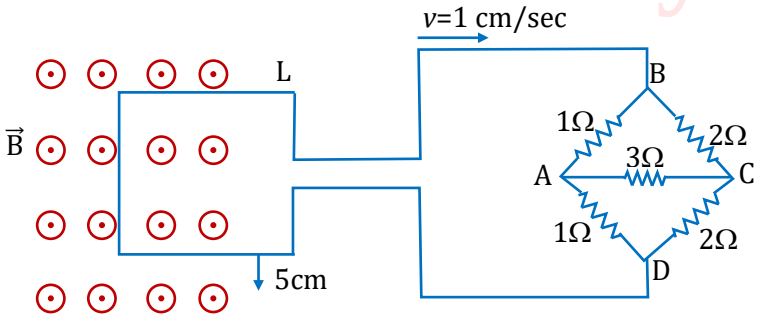
7. The total number of turns and cross-section area in a solenoid is fixed. However, its length L is varied by adjusting the separation between windings. The inductance of solenoid will be proportional to :

(A) $1/L^2$ (B) $1/L$ (C) L (D) L^2

PEM199

8. The figure shows a square loop L of side 5 cm which is connected to a network of resistances. The whole setup is moving towards right with a constant speed of 1 cms^{-1}. At some instant, a part of L is in a uniform magnetic field of $1T$, perpendicular to the plane of the loop. If the resistance of L is 1.7 Ω, the current in the loop at that instant will be close to :

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(A) 115 μA (B) 170 μA (C) 60 μA (D) 150 μA

PEM200

ANSWER KEY

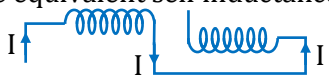
Question	1	2	3	4	5	6	7	8
Answer	D	C	D	A	A	C	B	B

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RACE
Regular Analysis Through Continuous Exercise

Electromagnetic Induction (EMI) : RACE-03

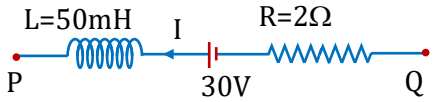
1. Two coils of self inductance L_1 and L_2 are connected in series combination having mutual inductance of the coils as M. The equivalent self inductance of the combination will be :



(A) $\frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{M}$ (B) $L_1 + L_2 + M$ (C) $L_1 + L_2 + 2M$ (D) $L_1 + L_2 - 2M$

PEM201

2. A part of a complete circuit is shown in the figure. At some instant, the value of current I is $1A$ and it is decreasing at a rate of $10^2 A\ s^{-1}$. The value of the potential difference $V_P - V_Q$, (in volts) at that instant, is.

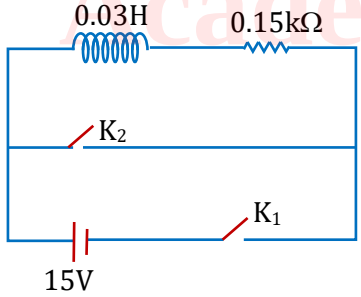


(A) 32 (B) 33 (C) 44 (D) 34

PEM202

3. An inductor ($L = 0.03\ H$) and a resistor ($R = 0.15\ k\Omega$) are connected in series to a battery of $15V$ EMF in a circuit shown below. The key K_1 has been kept closed for a long time. Then at $t = 0$, K_1 is opened and key K_2 is closed simultaneously. At $t = 1ms$, the current in the circuit will be ($e^5 \cong 150$)

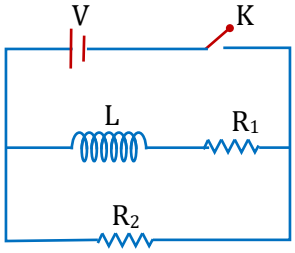
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(A) 6.7 mA (B) 0.67 mA (C) 100 mA (D) 67 mA

PEM203

4. In the circuit show below, the key K is closed at $t = 0$. The current through the battery is :



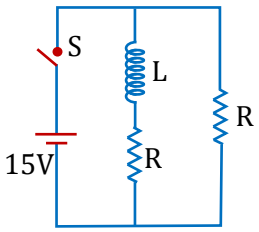
(A) $\frac{V(R_1+R_2)}{R_1R_2}$ at $t = 0$ and $\frac{V}{R_2}$ at $t = \infty$ (B) $\frac{VR_1R_2}{\sqrt{R_1^2+R_2^2}}$ at $t = 0$ and $\frac{V}{R_2}$ at $t = \infty$

(C) $\frac{V}{R_2}$ at $t = 0$ and $\frac{V(R_1+R_2)}{R_1R_2}$ at $t = \infty$ (D) $\frac{V}{R_2}$ at $t = 0$ and $\frac{VR_1R_2}{\sqrt{R_1^2+R_2^2}}$ at $t = \infty$

PEM204

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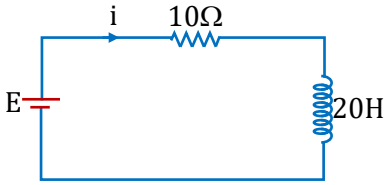
5. In the figure shown, a circuit contains two identical resistors with resistance $R = 5\Omega$ and an inductance with $L = 2mH$. An ideal battery of 15 V is connected in the circuit. What will be the current through the battery long after the switch is closed?



(A) $6A$ (B) $7.5A$ (C) $5.5A$ (D) $3A$

PEM205

6. A 20 Henry inductor coil is connected to a 10 ohm resistance in series as shown in figure. The time at which rate of dissipation of energy (joule's heat) across resistance is equal to the rate at which magnetic energy is stored in the inductor is :

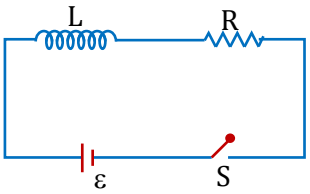


(A) $\frac{2}{\ell n2}$ (B) ℓn2 (C) 2ℓn2 (D) $\frac{1}{2}\ell n\, 2$

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PEM206

7. A shown in the figure, a battery of emf ε is connected to an inductor L and resistance R in series. The switch is closed at $t = 0$. The total charge that flows from the battery, between $t = 0$ and $t = t_c$ (t_c is the time constant of the circuit) is :



(A) $\frac{\varepsilon L}{R^2}\left(1-\frac{1}{e}\right)$ (B) $\frac{\varepsilon R}{eL^2}$ (C) $\frac{\varepsilon L}{R^2}$ (D) $\frac{\varepsilon L}{eR^2}$

PEM207

8. A coil of self inductance 10 mH and resistance 0.1 Ω is connected through a switch to a battery of internal resistance 0.9 Ω. After the switch is closed, the time taken for the current to attain 80% of the saturation value is : (Take ln5 = 1.6)

(A) 0.103s (B) 0.016s (C) 0.002s (D) 0.324s

PEM208

9. A coil of resistance 4Ω is connected across a 0.4 V battery. The current in the coil is 63 mA . 1 sec after the battery is connected. Find the inductance of the coil. [$e^{-1} = 0.37$]

(A) 2 H (B) 3 H (C) 4 H (D) 5 H

PEM209

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10. A power transmission lines feeds input power at 2300 V to a step down transformer with its primary windings having 4000 turns, giving the output power at 230 V. If the current in the primary of the transformer is 5A, and its efficiency is 90%, the output current would be :-

(A) 50 A (B) 25 A (C) 45 A (D) 20 A

PEM210

11. The primary of a 1 : 3 step - up transformer is connected to a source and the secondary is connected to a resistor R. The power dissipated by R in this situation is P. If R is connected directly to the source it will dissipate a power of :

(A) $P/9$ (B) $P/3$ (C) P (D) $3P$

PEM211

12. The primary and secondary coils of a transformer have 50 and 1500 turns respectively. If the magnetic flux ϕ linked with the primary coil is given by $\phi = \phi_0 + 4t$, where ϕ is in webers, t is time in seconds and ϕ_0 is a constant, the output voltage across the secondary coil is :–

(A) 30 volts (B) 90 volts (C) 120 volts (D) 220 volts

PEM212

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	B	C	A	C	D	B	C	C
Question	11	12								
Answer	A	C								

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